

ERD Review Questions – Answer Key and Explanations

1. Which of these is an entity?

■ b) Customer

An entity represents a real-world object or concept the organization stores data about. “Customer” fits this definition, while “Name” and “Age” are attributes of an entity.

2. Entities in ERDs are represented as:

■ c) Rectangles

Entities are drawn as rectangles, attributes as ovals, and relationships as diamonds.

3. True/False: An entity must always have a unique identifier.

■ True

Every entity type should have a unique identifier (primary key) to distinguish one instance from another.

4. Which is a derived attribute?

■ b) Age

Age can be derived from another stored value such as Date of Birth (DOB), so it's a derived attribute.

5. Composite attributes can:

■ a) Be split into subparts

A composite attribute (e.g., Full Name) is made up of smaller components (e.g., First Name, Last Name).

6. Attributes are shown in ERDs as:

■ a) Ovals

Attributes are represented by ovals connected to their entities or relationships.

7. Which shape represents relationships?

■ b) Diamond

Relationships, which link entities, are depicted using diamonds in ER diagrams.

8. A Student enrolls in Course → 'enrolls in' is:

■ b) Relationship

“Enrolls in” describes an action linking two entities (Student and Course), hence it is a relationship.

9. True/False: Relationships always have attributes.

■ False

Relationships can have attributes (like “Date” in an Enrollment relationship), but not all relationships do.

10. Which is 1:M?

■ **b) Customer–Orders**

One customer can place many orders, but each order belongs to only one customer — a one-to-many (1:M) relationship.

11. Which requires an associative entity?

■ **c) M:N**

Many-to-many relationships (e.g., Students–Courses) need an associative entity to resolve them into two 1:M relationships.

12. True/False: A teacher can teach multiple classes → 1:M.

■ **True**

One teacher can be linked to many classes, so the relationship is one-to-many.

13. Which is a primary key?

■ **a) StudentID**

A primary key uniquely identifies each record. StudentID serves that role, while Name or DOB may not be unique.

14. Which is a surrogate key?

■ **b) Auto-generated CustomerID**

A surrogate key is an artificial, system-generated identifier used when no natural key exists.

15. A foreign key is:

■ **b) A copy of another entity's PK**

A foreign key is an attribute in one entity that references the primary key of another to maintain referential integrity.